

name	axioms	$\vdash \forall x\varphi$: pure / over Q
H_V ; H_{sch}	$\{\forall x\varphi\}$; $\{\sigma_\varphi\}$, closed guard	yes/yes; no/no
H_{open} ; H_{both}	$\{\sigma_\varphi\}$, open guard ($z \sim \mathcal{Q}_{\text{open}}$); $\{\forall x\varphi, \sigma_\varphi\}$, weight w_V on $\forall x\varphi$	yes/yes; yes/yes
memorisers; over-specific	finite sets of closed instances; e.g. $\sigma_\varphi[z := Sz]$, closed guard	no/no; no/no
over-general	a subterm or subformula of σ_φ replaced by a fresh metavariable; the bare F_0 (inconsistent)	varies
root split	$\{\sigma_\varphi[z := f(z_1, \dots, z_r)] : f \text{ a root of } \mathcal{Q}\}$, closed guards	no/no
R_k	$\{\varphi(0), \dots, \varphi(S^{k-1}0), \varphi(S^k z)\}$, z unguarded; weights ($v_0, \dots, v_{k-1}, u$)	no/yes
Split_k ; Both0	$\{\varphi(0), \dots, \varphi(S^{k-1}0), \forall y\varphi(S^k y)\}$; $\{\varphi(0), \forall x\varphi\}$ (sentences only)	no/yes; yes/yes

Table 1: Hypotheses of Section 1^{universal §1.3}; provability for $\varphi = 0 + x = x$. A theory of closed instances proves $\forall x\varphi$ neither in pure logic nor over Q (AS Ex 4.9); R_k , Split_k : Propositions 1.22 and 1.24. $B \oplus_w A$ adds A with weight w to B , scaling B ’s weights by $1 - w$.

1 The case $\forall x\varphi$: what instances support

Hänni’s question starts from one case, statements $\varphi(t)$ and the law $\forall x\varphi$: “forall x phi(x) is a simple law which implies all the other statements. so once we see all these other statements, we should up the probability on forall x phi(x). i guess we should still be entertaining other hypotheses as well” (??). In the model of ?? the answer is this. The data favour the generator that emits exactly the instances, and never favour $\forall x\varphi$ over its instance schema σ_φ . “All future data are instances” is confirmed; “the axioms prove $\forall x\varphi$ ” is not, a Bayesian form of the ω -gap of AS §4.5. $\forall x\varphi$ is licensed by an instance at a parameter, by a quantified datum, or by a likelihood that models which theorems are reported. All results are from the universal track; proofs and further tables are in Appendix A. Logarithms are natural; rates are in nats per datum.

1.1 Setup

Definition 1.1 (Instances, schema, term laws). ^{universal §1.1–1.3} $\varphi(x)$ is an L_A -formula with exactly one free variable x , which occurs in it; the examples $(0 + x = x, x + 0 = x, x \cdot 0 = 0, \neg Sx = 0, x < Sx)$ are quantifier-free unless stated. $I_c(\varphi) := \{\varphi(t) : t \text{ closed}\}$; $I_o(\varphi)$ also allows parameters in t . The instance schema is $\sigma_\varphi := \varphi[z/x]$, z a 0-ary term metavariable, closed guard unless stated. Term laws: $\mathcal{Q}_{\text{num}}(S^j 0) = (1 - q)q^j$, $q = \frac{1}{2}$ by default; $\mathcal{Q}_{\text{GW}}$, each node picking its root $0, S, +, \cdot$ with probabilities .4, .3, .2, .1; $\mathcal{Q}_{\text{open}} := \rho[p] + (1 - \rho)\mathcal{Q}_{\text{num}}$ for a parameter p . The first two are root-generated: $\mathcal{Q}(f(t_1, \dots, t_r)) = p_f \prod_i \mathcal{Q}(t_i)$. One law supplies every metavariable body and every $\forall E$ term.

Calculi and likelihoods are as in ?????. In C_{min} each chain node cites (probability $1 - c$) or applies $\forall E$ with $t \sim \mathcal{Q}$ (probability c); $C_{\text{open}}(c, g)$ adds Gen on the parameter (g ; cite $p_{\text{cite}} = 1 - c - g$); μ_T is the unnormalised derivation mass and $L1 = \mu_T/Z_T$. Hänni’s scores are S_{prove} , S_{nc} and S_g with $g \equiv 1$ on provable data (the track’s $S_{\text{nc}, \beta}$ is $\beta^n S_g$ with $g_\infty = 1/\beta$; ??). Theorems about countable classes assume a proper prior (??); the computations use finite or summable families with the track’s prior $2^{-(\sum_A |A| + \#\text{axioms})}$, improper over all theories, and memorisers with a product-Bernoulli prior over a named universe ($r_s = 2^{-(|s|+1)}$, $\sum r_s < \infty$). Defaults: $c = 0.3$; $(c, g, \rho, q) = (0.3, 0.2, 0.1, \frac{1}{2})$ in C_{open} ; Dirichlet $\alpha = 1$ in checks c2–c7, $\alpha = \frac{1}{2}$ in c8–c10.

likelihood	O_n/O_0
L0 (strict citation)	0 for $n \geq 1$
$L0^{cl}; L1^{sel}(S)$ with $\forall x\varphi \notin S; S_{\text{prove}}, S_{\text{nc}}, S_g (\forall x\varphi$ consistent)	1
μ_T (unnormalised); $L1^{\max}$	c^n
L1	$(cZ_{H_{\text{sch}}}/((1-c)+cZ_{H_{\text{sch}}}))^n; (c/(1+c))^n$ for quantifier-free φ
background B , under μ_T	$\prod_i r(X_i), r(s) \in [c, 1]; r(s) = c$ if $\mu_B(s) = 0$

Table 2: Odds of $H_{\forall}$ against H_{sch} after n closed instances, relative to the prior odds (Theorem 1.3); asserted on every sampled path (5 formulas, 2 term laws, 50 seeds, $n \leq 200$), referee’s code to $7 \cdot 10^{-13}$. The consistency condition in the score row is ours; the notes leave it implicit.

1.2 Instance data never favour $\forall x\varphi$ over its schema

Proposition 1.2 (Factorisation). *[proved; computed]^{universal Prop U1} In $\mathbf{C}_{\min}$, let φ have one free variable and m leading quantifiers. Then $\mu_{H_{\forall}}(s) = c\mu_{H_{\text{sch}}}(s)$ for every sentence $s \neq \forall x\varphi$; $\mu_{H_{\forall}}(\forall x\varphi) = 1 - c$, $\mu_{H_{\text{sch}}}(\forall x\varphi) = 0$; $Z_{H_{\text{sch}}} = (1 - c)\sum_{k \leq m} c^k$, $Z_{H_{\forall}} = (1 - c) + cZ_{H_{\text{sch}}}$. With a background, $\mu_{B \oplus_w \forall x\varphi} = (1 - w)\mu_B + w\mu_{H_{\text{sch}}}$ and $\mu_{B \oplus_w \sigma_{\varphi}} = (1 - w)\mu_B + w\mu_{H_{\text{sch}}}$ on $s \neq \forall x\varphi$. For k variables the factor is c^k on each full instance.*

Proof idea. “Cite $\forall x\varphi$, eliminate with $t_1, \dots, t_k$ ” maps to “cite σ_{φ} with $z := t_1$, eliminate with $t_2, \dots, t_k$ ”: same conclusion, probability larger by $1/c$, a bijection onto all σ_{φ} -chains (Appendix A.1). Checked to $5 \cdot 10^{-14}$ on $2 \cdot 10^5$ sampled derivations.

Theorem 1.3 (Odds on instance data). *[proved; computed]^{universal Thm U2} In $\mathbf{C}_{\min}$, after n closed instances the odds $O_n := \pi_n(H_{\forall})/\pi_n(H_{\text{sch}})$ satisfy O_n/O_0 as in Table 2.*

Theorem 1.4 (Instance data never favour $\forall x\varphi$). *[proved]^{universal Thm B} On data not containing $\forall x\varphi$, the posterior odds of the $\forall$ -version against the schema version are at most the prior odds: (B1) in $\mathbf{C}_{\min}$, single axioms, closed instances, under $L0^{cl}$, μ_T , L1, $L1^{sel}$, $L1^{\max}$ and the scores (equality under $L0^{cl}$, $L1^{sel}$, scores); (B2) in $\mathbf{C}_{\min}$, $B \oplus_w \forall x\varphi$ against $B \oplus_w \sigma_{\varphi}$ for any background B and any data, under μ_T and L1, with a fixed w or a common weight prior (equality under $L0^{cl}$); (B3) in $\mathbf{C}_{U3}$ calculi, quantifier-free φ , under μ_T , $L1^{\max}$ and the scores.*

Proof idea. (B2) holds at every w , and $Z_{H_{\forall}} - Z_{H_{\text{sch}}} = (1 - c)(1 - Z_{H_{\text{sch}}}) \geq 0$ orders the normalisers. (B3) rests on a simulation inequality: in $\mathbf{C}_{U3}$ every citation of $\forall x\varphi$ sits under a $\forall E$, and replacing the pair by a citation of σ_{φ} divides the probability by c (Proposition A.1). The mechanism is the size principle (??): $\forall x\varphi$ spends mass on outputs never observed. Not covered: $L1^{sel}$ with a background and learned weights (Proposition 1.7), $\wedge$ -detours (Remark 1.6), normalised likelihoods in $\mathbf{C}_{U3}$.

Proposition 1.5 (Robustness of the direction). *[proved; computed]^{universal Props U2n, U2t, U2c} In $\mathbf{C}_{\min}$, for every $s \neq \forall x\varphi$: (U2n) with a full-support noise law N and $\eta \in (0, 1)$, the ratio of the noisy laws of $H_{\forall}$ and H_{sch} at s is in $[c, 1]$ under μ_T , ≤ 1 under L1, and 1 under $L1^{sel}$ with noise after selection; a false datum no longer refutes. (U2t) If chains are weighted by $f(\ell)w_A \prod_j Q(t_j)$, f non-increasing in the number ℓ of eliminations (e.g. Levin-style $f(\ell) \propto (\ell + 1)^{-2}$), then $P_{H_{\forall}}^f(s) \leq \sup_{\ell} \frac{f(\ell+1)}{f(\ell)} P_{H_{\text{sch}}}^f(s) \leq P_{H_{\text{sch}}}^f(s)$. (U2c) Under L1, quantifier-free φ and any prior on c , $O_n/O_0 \leq 2^{-n}$.*

Remark 1.6 ($\wedge$ -detours). [refuted (conjecture); counterexample: proof sketch; computed]^{universal §2.3} The first version conjectured that $\wedge$ -detours cannot break the factor c of Proposition A.1. They can: in $\mathbf{C}_\wedge$ with $(p_{\text{cite}}, c, a, e) = (0.35, 0.15, 0.25, 0.25)$ and $\varphi = 0 + x = x$, $\mu_{H_\forall}(\varphi(t))/\mu_{H_{\text{sch}}}(\varphi(t)) = 0.15553 > c$ (Catalan spine formula, Appendix A.1; samplers 0.15528 ± 0.00067 and 0.1562 ± 0.0005). On a 9139-point grid the ratio exceeds c but stays ≤ 0.925 (normalised ≤ 0.48). The factor c is specific to $\mathbf{C}_{\min}$ and $\mathbf{C}_{\text{U3}}$; whether detours can reverse the direction is open.

Proposition 1.7 (Learned weights under $\mathbf{L1}^{\text{sel}}$). [proved; computed]^{universal Prop B2} In $\mathbf{C}_{\min}$ let $B = \{\sigma_\psi\}$, ψ quantifier-free, $\text{inst}(\sigma_\psi) \cap \text{inst}(\sigma_\varphi) = \emptyset$, and compare $B \oplus_w \forall x \varphi$ with $B \oplus_w \sigma_\varphi$ (uniform prior on w , equal prior masses) under $\mathbf{L1}^{\text{sel}}$, S the closed quantifier-free sentences. On data i.i.d. from $(1-w^*)\mathcal{Q}_{\sigma_\psi} + w^*\mathcal{Q}_{\sigma_\varphi}$ the Bayes factor tends a.s. to $h(w^*)$, $h(v) := c/(c+v(1-c))^2 \in [c, 1/c]$, which exceeds 1 iff $w^* < \sqrt{c}/(1+\sqrt{c})$.

The cause is the induced prior on the effective weight $wc/(1-w+wc)$ (computed: within $4 \cdot 10^{-4}$ of $h(w^*)$ at $n = 10^5$). The effect is bounded and is not evidence that the axioms prove $\forall x \varphi$.

Remark 1.8 (Per-citation selection). [proved]^{editor, from experiments §1.7, Prop X6(b) and universal Prop B2} Under the per-citation $\mathbf{L1}_{\text{cit}}^{\text{sel}}$ both theories of Proposition 1.7 have law $(1-w)\mathcal{Q}_{\sigma_\psi} + w\mathcal{Q}_{\sigma_\varphi}$ at every w , since the selected law of the component $\forall x \varphi$ is $\mathcal{Q}_{\sigma_\varphi}$ (experiments Prop X6(b)): the Bayes factor is 1 at every n (cf. ??).

Proposition 1.9 (Spare-slot prover). [proved; computed]^{universal §4} In $\{H_{\text{sch}}, H_\forall, H_{\text{both}}\}$ with a uniform prior on $w_\forall$, under $\mathbf{L1}$ with quantifier-free φ , the likelihood ratio $H_{\text{both}} : H_{\text{sch}}$ is $I_n = \int_0^1 \left(\frac{1-w(1-c)}{1+wc}\right)^n dw \in [\frac{1}{n+1}, \frac{1}{n(1-c)}]$, so $P(T \vdash \forall x \varphi \mid D_n) \rightarrow 0$ like $\Theta(1/n)$: a spare slot (??).

The experiments reproduce Theorem 1.3 in their chain Ch_1 , whose factor $c_{\text{ch}} = \frac{1}{2}$ matches the unnormalised row: one bit per datum to $1.1 \cdot 10^{-11}$, and under $\mathbf{L1}_{\text{cit}}^{\text{sel}}$ an exact tie with $P(T \vdash \forall x \varphi \mid D_n) \rightarrow 0.484$ (0.269 for $\neg Sx = 0$), the prior share in their code (E1, ??).

1.3 Memorisers, over-general and over-specific templates

Theorem 1.10 (Size principle on instances). [proved; computed]^{universal Thm U4} Let the data be i.i.d. from P^* , $P^*(I) = 1$ for $I = I_c(\varphi)$, and P' a sub-probability with $P'(I) = p < 1$. Then $\text{KL}(P^* \| P') \geq \ln \frac{1}{p}$, $\frac{1}{n} \ln \frac{P'(D_n)}{P^*(D_n)} \rightarrow -\text{KL}(P^* \| P')$ a.s., and $\mathbb{E} \frac{P'(D_n)}{P^*(D_n)} \leq p^n$. A first-order template strictly more general than σ_φ , with metavariables i.i.d. from $\mathcal{Q}$ and a formula law F , has $P_\tau(I) \leq \max(\max_f p_f, \max_g F(\text{root} = g))$.

Proposition 1.11 (Over-specific templates). [proved; computed]^{universal Prop U5} For $\tau = \sigma_\varphi[z := f(z_1, \dots, z_r)]$ and root-generated $\mathcal{Q}$, $P_\tau(\varphi(t))/P_{H_{\text{sch}}}(\varphi(t))$ is $1/p_f$ if $\text{root}(t) = f$ and 0 otherwise. So the odds $\tau : \sigma_\varphi$ grow like p_f^{-n} while all data have root f (probability p_f^n), then vanish; they form a martingale. Some single-root template survives with probability $\sum_f p_f^n$, the failure probability of AS Thm 4.4(a).

Proposition 1.12 (Memorisers). [(i), (ii), (ii') proved; (iii) conjecture; computed]^{universal Prop U6} Let $Z_0 := \prod_s (1-r_s) > 0$, S_n the distinct data, $m_n = |S_n|$, μ the prior of the memoriser class. (i) The class marginal lies in $[Z_0, 1] \cdot \prod_{s \in S_n} r_s m_n^{-n}$ for uniform weights ($\text{DirMult}_\alpha(D_n)$ in place of m_n^{-n} for Dirichlet weights). (ii) Uniform weights, $\mathcal{Q}$ of infinite support and finite entropy: the log-odds against H_{sch} divided by n tend to $-\infty$ a.s. (ii') Laplace weights: deterministically the log-odds are $\geq \ln \frac{\mu Z_0}{\pi(H_{\text{sch}})} + \sum_{s \in S_n} \ln r_s - (m_n - 1) \ln(n + m_n - 1)$, which under $\mathcal{Q}_{\text{num}}$ (numeral universe) is $\geq -C \ln^2 n$ a.s. for all large n : the class is not exponentially small. (iii) Conjecture: $-\Theta(\ln^2 n)$ under $\mathcal{Q}_{\text{num}}$.

So the brief’s H4(a), “mass leaves memorisation and over-general templates exponentially”, is proved for over-general templates and refuted for memorisation in general; under $\mathcal{Q}_{\text{GW}}$ the computed rate is exponential, as distinct data grow linearly (Appendix A.2). E1 tests only the single memoriser $\text{Mem}(D_n)$; every fixed memoriser dies under well-specification (??).

1.4 Predictive confirmation

Theorem 1.13 (Predictive confirmation). *[proved; computed]^{universal Thm U8} Let $M(\cdot \mid D_n) = \sum_T \pi_n(T)P_T$, $G := \{T : P_T(I) = 1\}$, $T^* \in G$. (1) $\sum_{n \geq 0} \mathbb{E}[1 - M(I \mid D_n)] \leq \ln \frac{1}{\pi(T^*)}$. (2) The probability that all future data lie in I is $\pi_n(G)$, and $\pi_n(G) \rightarrow 1$ a.s. (3) $\mathbb{E}[1 - \pi_n(G)] \leq \min(1, \sum_{T \notin G} \frac{\pi(T)}{\pi(T^*)} P_T(\text{supp } P_{T^*})^n)$. (4) If $T \in G$ have law $\hat{P}$ and $T \notin G$ have law $(1 - \varepsilon_T)\hat{P} + \varepsilon_T N$, $N(I) = 0$, then on instance sequences $1 - \pi_n(G) = \frac{\sum_{T \notin G} \pi(T)(1 - \varepsilon_T)^n}{\pi(G) + \sum_{T \notin G} \pi(T)(1 - \varepsilon_T)^n}$.*

The rate depends on near-universal competitors: $\approx (1 - \pi_0)/(\pi_0 n)$ for Laplace’s rule with a point mass π_0 , but $1 - \pi_n(G) = 0.37$ at $n = 10^{50}$ for a dyadic family ($\varepsilon_j = 2^{-j}$, $\pi_0 = 0.01$).

Remark 1.14 (Relation to Hutter). *[known; computed]^{universal §5} In the Bayes–Laplace model the universal hypothesis, even its observable form, has posterior 0, while the universal prior confirms it (Hutter, 2007) (abstract checked; full text not). Theorem 1.13(2) is the analogue for countable classes, by the same mechanism: prior mass on instance-only hypotheses. But G contains H_{sch} , which does not prove $\forall x\varphi$. Nor is confirmation stepwise monotone: with prior $\frac{1}{2}$ on each of σ_φ , $\varphi(Sz)$, the datum $\varphi(S0)$ lowers σ_φ to $\frac{1}{3}$, a Nicod-type effect (Leike and Hutter, 2015).*

1.5 The Bayesian ω -gap

Theorem 1.15 (Bayesian ω -gap). *[(a), (b), (c1)–(c3), (d1), (e) proved; (d2) proved for the pair, proof sketch in general; computed]^{universal Thm U10} Let $\mathcal{H}$ be countable with a proper prior $\pi > 0$, $\text{Prov}_\forall := \{T \in \mathcal{H} : T \vdash \forall x\varphi\}$ (first-order; inconsistent theories included), and the data i.i.d. from P_{T^*} , $T^* \in \mathcal{H}$, unless stated.*

- (a) $P(T \vdash \forall x\varphi \mid D_n) \rightarrow \pi(\text{Prov}_\forall \cap C^*)/\pi(C^*)$ a.s.; likewise for $\vdash_{\mathcal{C}}$ and for Bel, Dis, Indep, Inc (??).
- (b) Prior share. If C^* has a prover and a non-prover, the limit is in $(0, 1)$: $H_\forall, H_{\text{sch}}$ under L0^{cl} or L1^{sel} ; $B \oplus \forall x\varphi$, $B \oplus \sigma_\varphi$ under L0^{cl} with a common weight or weight prior. In a class of the two theories of Proposition 1.7 under L1^{sel} the laws differ, but that proposition gives a limit in $(0, 1)$ directly.
- (c) Faithful regime. Call a likelihood faithful for $\mathcal{C}$ if $\text{supp } P_T = \text{Th}_{\mathcal{C}}(T) \cap X$ for all T (X the data space). (c1) If $\forall x\varphi \in X$, $P(T \vdash_{\mathcal{C}} \forall x\varphi \mid D_n) \rightarrow 1[T^* \vdash_{\mathcal{C}} \forall x\varphi]$. (c2) If also $\mathcal{C}$ is sound and all axiom instances lie in X , members of C^* have T^* ’s first-order theorems and $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 1[T^* \vdash \forall x\varphi]$. (c3) In $\mathcal{C}_{\text{min}}$, and $\mathcal{C}_{\text{open}}$ with a guard, L1 with full-support $\mathcal{Q}$ is faithful and $\text{supp } P_{H_{\text{sch}}} = I_{\mathcal{C}}(\varphi)$ for quantifier-free φ , so for $T^* = H_{\text{sch}}$ the limit in (c1) is 0. If $\mathcal{C}$ has a rule that maps an instance to a non-instance with positive probability ($\wedge I$, $\vee I$, symmetry of $=$, $\rightarrow I$, MP with cited logical axioms), every T with $P_T(I_{\mathcal{C}}) > 0$ emits non-instances: instance-only data are then misspecified for every hypothesis, and the limit is set by the best-fitting generators of the class (??).
- (d) Filtered data from $H_\forall$, seen only in $S = I_{\mathcal{C}}(\varphi)$. (d1) For quantifier-free φ in $\mathcal{C}_{\text{min}}$ the filtered law is H_{sch} ’s: limit 0 under L1 , the prior share under L1^{sel} . (d2) For quantified φ , H_{sch} is

the KL-minimiser in $\{H_\forall, H_{\text{sch}}\}$ and the limit is 0; for larger classes the minimiser was not determined.

- (e) Unknown filter. With a countable filter family $\mathcal{F}$, prior ν independent of π , and pair laws $P_T(s)1[s \in S]/P_T(S)$, (a) applies to pairs. For $\{H_\forall, H_{\text{sch}}\}$, $\mathcal{C}_{\text{min}}$, quantifier-free φ and data law $\mathcal{Q}_{\sigma_\varphi}$ the limit is $\frac{\pi(H_\forall)\nu(\mathcal{F}_1)}{\pi(H_\forall)\nu(\mathcal{F}_1) + \pi(H_{\text{sch}})\nu(\mathcal{F}_0)}$, $\mathcal{F}_0 = \{S \supseteq I_c\}$, $\mathcal{F}_1 = \{S \in \mathcal{F}_0 : \forall x \varphi \notin S\}$.

Proof idea. (a) is ?? summed over $\text{Prov}_\forall$; the rest identifies C^* in each setting (Appendix A.4). Computed: on closed data in $\mathcal{C}_{\text{open}}$ the prover mass falls from 0.750 to 0.004 at $n = 50$ under L1 and stays 0.750 under L1^{sel} (Table 4); (e) reaches its predicted 0.2500 by $n = 10$.

In the prior-share regime the answer is the prior share. That it lies in $(0, 1)$ is robust; its value is not: for $\varphi = 0 + x = x$ the share of $H_\forall$ in $\{H_\forall, H_{\text{sch}}\}$ is 0.333 in the universal code and in π_1 of ??, 0.200 in π_2 , 0.484 in the experiments' code and 0.030–0.042 in pa's codes (Table 5). The data never move it.

Proposition 1.16 (Hänni's variants on $\forall x\varphi$). *[proved; computed]^{universal §6.2} Let $\varphi = 0 + x = x$, $\mathcal{Q}$ of full support. Under S_{prove} the posterior converges a.s. in total variation to the prior restricted to $\{T \text{ consistent} : T \vdash I_c(\varphi)\}$, which contains $\mathbf{Q}$ (independent), $\mathbf{Q} + \forall x\varphi$ (true), $\mathbf{Q} + \exists x\neg\varphi$ (false), H_{sch} , $H_\forall$ and consistent over-general templates such as $\{0 + z_1 = z_2\}$: all three trichotomy values keep positive mass, and over-generalisation is never penalised. The same holds for S_{nc} and S_g ($g \equiv 1$ on provable data). Under generative likelihoods inconsistent theories die within a few data (Table 6).*

Inconsistent theories inflate Bel and Dis alike (at $n = 0$ in the class of Table 3, $\text{Bel}(\forall x\varphi) \approx 0.69$ is mostly Inc), so the verifier should use Bel_{cons} (??).

Proposition 1.17 (An ω -gap for $\mathbf{Q4}$). *[proved; computed]^{universal Prop U13} ($\mathbf{Q} - \mathbf{Q4}$) plus all closed instances of $x + 0 = x$ does not prove $\forall x(x + 0 = x)$ (a model on $\mathbb{N} \cup \{a, b\}$). So closed instances cannot tell the axiom $\mathbf{Q4}$ from its closed schema $z + 0 = z$, which differ in their theorems over $\mathbf{Q} - \mathbf{Q4}$.*

Proposition 1.18 (The Gaifman condition). *[proved; known]^{universal Prop U15} Let μ be a probability on L -structures, $P(\psi) := \mu\{M : M \models \psi\}$, and $t_1, t_2, \dots$ the closed terms. Then*

$$P(\forall x\varphi \mid \bigwedge_{i \leq n} \varphi(t_i)) \rightarrow P(\forall x\varphi) / \mu\{M : M \models \varphi(t) \text{ for all closed } t\}.$$

The Gaifman condition (Gaifman, 1964) $P(\forall x\varphi) = \inf_n P(\bigwedge_{i \leq n} \varphi(t_i))$ holds for all φ iff $M_0 \preceq M$ for μ -a.e. M , with M_0 the closed-term substructure (AS Prop 4.7); the limit is then 1 if $P(\forall x\varphi) > 0$ and 0 if $P(\forall x\varphi) = 0$.

So believing $\forall x\varphi$ from true closed instances is safe for $\mathbb{N}$ and unsafe for credences that weigh $\mathbb{R}$, V , models of $\text{PA} + \neg\text{Con}(\text{PA})$ (AS Ex 4.8) or some models of $\mathbf{Q}$ (AS Ex 4.9). The condition is cited from memory by the track; secondary sources state it with constants rather than closed terms.

1.6 Open instances and the closedness guard

Proposition 1.19 (Parameter data and the guard). *[proved; computed]^{universal Prop U11(a)–(d)} Assume the closure reading, Gen over parameters, theories without parameters, and a likelihood supported on theorems. (a) Once D contains some $\varphi(p)$, every T with $\pi(T \mid D) > 0$ proves $\forall x\varphi$. (b) If $P_{T^*}(\varphi(p)) = \beta > 0$, then $P(P(T \vdash \forall x\varphi \mid D_n) < 1) \leq (1 - \beta)^n$. (c) In $\mathcal{C}_{\text{open}}(c, g)$ with $\mathcal{Q}_{\text{open}}$, per*

closed datum under L1: $H_{\text{open}} : H_{\text{sch}} = \frac{1-\rho}{1+g\rho}$, $H_{\forall} : H_{\text{sch}} = \frac{c(1-\rho)}{1+c}$, $H_{\forall} : H_{\text{open}} = \frac{c(1+g\rho)}{1+c}$. (d) If the guarded H_{sch} is in the class, closed data give $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 0$: the guard is learned at rate $\mathcal{W}_0 := \ln \frac{1+g\rho}{1-\rho}$ (0.1252 at the defaults), and H_{both} decays like $1/n$.

Here ρ is fixed. In pa's two-part code, where the parameter probability is learned, the analogous head-symbol split of $x + 0 = x$ closes in on the schema at only about 5 bits per doubling of n (??); each rate holds in its model.

Remark 1.20 (Without a guard). [refuted; computed]^{universal §7.2} The first version claimed: “without a guard (DT^o, λ -convention, parameters admissible) H_{open} wins on closed data and every surviving hypothesis proves $\forall x\varphi$: the posterior takes the ω -step”. Counterexample: the unguarded DT^o pair $R_1 = \{\varphi(0), \varphi(Sz)\}$ beats H_{open} by $(1-q)\mathcal{W}_0$ per datum (Proposition 1.21) and does not prove $\forall x\varphi$ in pure logic; in $\{H_{\forall}, H_{\text{open}}, R_1\}$, $P(T \vdash \forall x\varphi \mid D_n)$ is 0.697 at $n = 100$ and 0.000 from $n = 1000$ (referee: 0.880, then 0.000 at 300). The claim survives for the class $\{H_{\forall}, H_{\text{open}}\}$ and under L1^{sel}. This is the Bayesian counterpart of AS Prop 4.5(b), (c).

Proposition 1.21 (Rates on closed data). [proved; computed]^{universal Prop N2} On closed numeral data from $\mathcal{Q}_{\text{num}}$, under L1: $\text{KL}(P^* \parallel P_{H_{\forall}}) = \ln \frac{1+c}{c(1-\rho)}$, $\text{KL}(P^* \parallel P_{H_{\text{open}}}) = \mathcal{W}_0$, $\inf_w \text{KL}(P^* \parallel P_{R_k}) = q^k \mathcal{W}_0$. Each split level halves the loss at $q = \frac{1}{2}$ (0.0626, 0.0313, 0.0156 for R_1, R_2, R_3).

The proof uses the exact law of R_k (Proposition A.3) and a waste lemma (Lemma A.2): a law that reproduces P^* on a tail of mass τ and wastes a fixed fraction r_w of the tail's mass loses at best $\tau \ln(1+r_w)$; for R_k , $\tau = q^k$ and $\ln(1+r_w) = \mathcal{W}_0$.

Proposition 1.22 (Who wins without a guard). [provability, (a), (b) proved; (c) proof sketch; computed; (d) computed, limit conjecture]^{universal Prop N3} Closed data i.i.d. from $\mathcal{Q}_{\text{num}}$; the class contains $H_{\forall}$, H_{open} , R_k for k in a nonempty $\mathcal{K}$ (learned or optimal weights), possibly memorisers. In pure logic $H_{\forall}$, H_{open} prove $\forall x\varphi$, and R_k and memorisers do not; over $\mathcal{Q}$ all but the memorisers do. (a) L1, pure logic, provers only $H_{\forall}, H_{\text{open}}$: $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 0$ a.s., exponentially, at rate $\geq (1-q^k)\mathcal{W}_0$ for each $k \in \mathcal{K}$. (b) L1, over $\mathcal{Q}$, $\mathcal{K}$ finite: $\rightarrow 0$ with Laplace-weighted memorisers, = 1 at every n without them. (c) L1^{sel}, finite $\mathcal{K}$: $\rightarrow 1$ in pure logic. (d) L1, $\mathcal{K} = \{1, \dots, 40\}$, memorisers: 1.000 over $\mathcal{Q}$ from $n = 10$ to 10^{12} , with ever deeper splits.

Proof idea. R_k fails $\forall x\varphi$ in $\mathbb{N} \cup \{e, e'\}$ with $Se = Se' = e'$, $0 + e = 0 + e' = e'$; over $\mathcal{Q}$, k uses of Q3 reduce $\forall x\varphi$ to closed instances and $\forall y\varphi(S^k y)$. (a), (b) compare the linear rates of Proposition 1.21 with each other and with the $O(\ln^2 n)$ cost of memorisers; (c) uses the $\frac{k}{2} \ln n$ Occam term (Schwarz, 1978; Clarke and Barron, 1990), the almost-sure uniform step not written out.

In c8 (Table 7), with $R_{1..4}$ and memorisers the over- $\mathcal{Q}$ value is 1.000 up to $n = 10^4$ and 0.000 from 10^5 ; with $R_{1..40}$ it stays 1.000 to $n = 10^{12}$, the MAP moving from R_3 ($n = 10^3$) to R_{30} (10^{12}). So without a guard, under a likelihood that ignores the selection, the posterior does not take the ω -step in pure logic; it approximates the missing guard by ever deeper splits, whose waste $q^k \mathcal{W}_0$ tends to 0. Over $\mathcal{Q}$ the answer is a race between the split family (provers) and the memoriser family (non-provers): a property of the class. Experiments E3(a) show a related drift when the term law, not a filter, is misspecified: the best closed-guard split R_m deepens from $m = 9$ –10 at $n = 512$ to 11–12 at 1024 (??); these runs are too short to test (b).

1.7 Quantified data, lemmas and the equational fragment

Proposition 1.23 (Quantified data). [(a) proved; (b) $f_q = 0$ proved, $f_q > 0$ proof sketch; computed; (c) proof sketch]^{universal Prop U12} (a) If $s \in D$ and $B \cup I_c(\varphi) \not\models s$, then under any likelihood supported

on theorems $B \oplus \sigma_\varphi$ has posterior 0; over $\mathbf{Q}$, one datum $\forall y(0 + Sy = Sy)$ leaves only provers of $\forall x(0 + x = x)$. (b) If all theories prove $\forall x\varphi$ at different costs ($\{\forall^L \forall x\varphi, \sigma_\varphi\}$ against $\{\forall x\varphi, \sigma_\varphi\}$ in $\mathbf{C}_{\min}$, L vacuous quantifiers, uniform weight priors, equal prior masses, data $\forall x\varphi$ with probability f_q and instances otherwise), the log Bayes factor tends to $\ln(1 + c + \dots + c^L)$ when $f_q = 0$ (0.349 at $L = 3$) and grows linearly when $f_q > 0$, at a rate a little above $f_q L \ln \frac{1}{c}$. (c) In $\mathbf{L1}^{\max}$ form a derived sentence becomes an axiom when the derivation steps it saves, at $\geq \ln \frac{1}{c}$ each, exceed its prior cost and an $O(\ln n)$ dilution: frequently used lemmas become axioms.

So the posterior identifies the data producer’s working set of primitives, not a canonical axiomatisation (compare ??). The brief’s “statements given without proof should be easily derivable” was tested on the equational fragment $E = \{Q4, Q5\}$ (Proposition A.4): every derivation of $0 + S^k 0 = S^k 0$ has at least $k + 1$ axiom leaves, so the two-part ($\mathbf{L1}^{\max}$) comparison favours the schema $0 + z = z$ linearly in k at $q = \frac{1}{2}$, while the full-sum comparison is shown to do so only under the sufficient condition $q > e^{-(1-\mu)^2/2}$ (μ the mean number of premises), which needs $q > 0.61$ and says nothing at $q = \frac{1}{2}$. Full $\mathbf{Q}$, and the full sum at $q = \frac{1}{2}$, are open. A Kt-style penalty on generating $0 + S^k 0 = S^k 0$ from E would charge only $O(\ln k)$ (??).

1.8 Without templates

Hänni proposes templates against his collapse. Without them, hypotheses are finite sets of sentences and σ_φ is unavailable. Setting: $\mathbf{C}_{\min}$, $c = 0.3$, numeral data, $q = \frac{1}{2}$, $\varphi = 0 + x = x$, learned weights.

Proposition 1.24 (Sentence-only class). *[(a) proof sketch; computed; (b), (c) proved; (d) computed, limit conjecture]^{universal Prop U16} (a) Under $\mathbf{L0}^{\text{cl}}$, $H_\forall$ has the true law and Split_k , Both0 match it only at a point or at the boundary, so $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 1$ (Laplace expansion uniform in k not written out). (b) Under $\mathbf{L1}$: $\text{KL}(P^* \| P_{H_\forall}) = \ln \frac{1+c}{c}$ (1.466), $\inf_w \text{KL}(P^* \| P_{\text{Split}_k}) = q^k \ln \frac{1+c}{c}$, $\inf_w \text{KL}(P^* \| P_{\text{Both0}}) = q \ln \frac{1+qc}{qc}$ (1.018). (c) Split_k proves $\forall x\varphi$ over $\mathbf{Q}$, not in pure logic; memorisers in neither. (d) In pure logic $P(T \vdash \forall x\varphi \mid D_n) \rightarrow 0$ in every class computed. Over $\mathbf{Q}$, with depth ≤ 4 memorisers win (limit 0); with depth ≤ 40 splits win to $n = 10^{12}$ under four prior variants (Table 8). Conjecture: with all depths the limit is 1.*

Lemma 1.25 (Survivors prove $\forall x\varphi$ over $\mathbf{Q}$). *[proved]^{universal Lemma S1} In $\mathbf{C}_{\min}$ with numeral elimination terms and $\varphi = 0 + x = x$, every finite set T of sentences with $P_T(\varphi(S^j 0)) > 0$ for infinitely many j satisfies $\mathbf{Q} \cup T \vdash \forall x\varphi$.*

A fixed memoriser is eventually refuted but the memoriser family is not, so the over- $\mathbf{Q}$ limit is a race between families. The first version’s “= 1 over $\mathbf{Q}$ ” (a three-hypothesis class) and its referee’s “ $\rightarrow 0$ over $\mathbf{Q}$ ” (bounded depth) were class artefacts (Appendix A.7).

Remark 1.26 (Kreisel’s conjecture). *[known (secondary sources only)]^{universal §9} “All instances have short derivations, so $\forall x\varphi$ is derivable” resembles Kreisel’s conjecture, reported open for standard formalisations of PA, settled for some (Parikh, 1973; Baaz and Pudlák, 1993) and false for some theories close to PA (Hrubeš, 2007) (none of these was opened). Split_1 shows that the naive version fails without axioms such as $Q3$: every numeral instance has a one-step derivation, $\forall x\varphi$ none.*

1.9 The intuition, precisely

The intuition is right against hypotheses that fit worse (over-general templates, memorisers), for prediction (“all future data are instances”), and for truth under a Gaifman-type credence, safely

in $\mathbb{N}$. It needs refinement in three places. σ_φ is a second simple law that implies the data, and it is never overtaken (Theorem 1.4); templates, proposed against the collapse, are what bring it into the class. “The axioms prove $\forall x\varphi$ ” does not gain from closed instances (Theorem 1.15). A derivation likelihood rewards the most specific generator that fits, and frequently used lemmas, not the logically strongest law. What licenses $\forall x\varphi$ is an instance at a parameter, a quantified consequence, or a selection-aware likelihood without a guard (Proposition 1.19(a), Proposition 1.23(a), Proposition 1.22(c)), which takes the ω -step by design, soundly in $\mathbb{N}$ and unsoundly in $\mathbb{R}$. Data at parameters are common in proof-assistant practice, so the practical answer is often positive, for a reason other than the intuition’s.

A good version for this case: a proper template prior, inconsistent theories excluded or reported as Inc; a derivation likelihood that models the selection of reported data, by a known filter ($\mathbf{L1}^{\text{sel}}$) or a prior over filters (Theorem 1.15(e)); a noise component (Proposition 1.5); and the verifier “accept s iff $\text{Bel}_{\text{cons}}(s) \geq 1 - \delta$ ”. On closed instances it accepts $\forall x\varphi$ only if the prior share of provers over theories and filters is $\geq 1 - \delta$; one parameter instance, or one quantified datum entailing it over the background, suffices.

Corollary 1.27 (Soundness for $\forall x\varphi$). *[proved]^{universal §11} Let the data be i.i.d. from P_{T^*} with fixed weights, $T^* \in \mathcal{H}$, $T^* \not\models \forall x\varphi$, and $C_\forall^* := \{T \in C^* : T \vdash \forall x\varphi \Leftrightarrow T^* \vdash \forall x\varphi\}$. The probability that the verifier (on Bel or Bel_{cons}) ever accepts $\forall x\varphi$ is at most $\delta/\pi(C_\forall^*) \leq \delta/\pi(T^*)$, uniformly in time: ?? read at the sentence $\forall x\varphi$.*

Proof. Members of $C_\forall^*$ do not prove $\forall x\varphi$ and share T^* ’s likelihood, so

$$\pi_n(C_\forall^*) = \pi(C_\forall^*) P_{T^*}(D_n)/M(D_n).$$

Acceptance needs $\pi_n(C_\forall^*) \leq \delta$, that is, $M(D_n)/P_{T^*}(D_n) \geq \pi(C_\forall^*)/\delta$. This ratio is a nonnegative supermartingale started at 1; by Ville’s inequality it ever reaches that level with probability at most $\delta/\pi(C_\forall^*)$. $\square$

Without well-specification there is no guarantee. In the misspecified settings of Theorem 1.15(d) and of Propositions 1.22 and 1.24, the limit theories (σ_φ , R_k , Split_k , memorisers) are true in $\mathbb{N}$, so in pure logic the verifier errs towards incompleteness; misspecification can also make it accept a falsehood with probability 1 (????). The learned-weight computations here are outside the corollary; for them see ????

Remark 1.28 (Relation to the MDL finding). *[proved (from the cited results)]^{universal §10} AS §6.10 found that “MDL tracks the statistics of usage, not the logical boundaries of schemas”. For $\forall x\varphi$: well specified, a root split with weights p_f has exactly the schema’s law under $\mathbf{L0}$ and $\mathbf{L1}$ (??; it keeps its prior share 0.004 in Table 3), and with learned weights it pays a polynomial Occam factor (??; AS Prop E.17); an unmodelled selection creates linear split wins, at rates $(1 - q^k)\mathcal{W}_o$ for R_k and $(1 - q^k) \ln \frac{1+c}{c}$ for Split_k (the analogue of AS Prop E.16, split by numeral depth rather than main connective); modelling the selection removes them (Proposition 1.22(c)). So a derivation likelihood does not change the MDL finding; it adds a way to be misspecified, since the theorems humans state are a selection from those their axioms generate (??).*

A Proofs and computations for Section 1

Proofs follow the universal track’s final notes; steps they leave out are named. Checks are the seeded scripts c1–c10 in `research/tracks/universal/checks/`; the referee’s independent scripts r1–r7 (which do not import the track’s code) are in `referee_code/`. A second run of all scripts reproduced every output byte for byte. Dirichlet $\alpha = 1$ in c2–c7, $\alpha = \frac{1}{2}$ in c8–c10.

A.1 Factorisation, odds and robustness

Proof of Proposition 1.2. A $C_{\min}$ -derivation from $\{\forall x\varphi\}$ is a chain: cite $\forall x\varphi$, then $k \geq 0$ eliminations with terms $t_1, \dots, t_k$, with probability $c^k(1-c) \prod_j Q(t_j)$. With $k = 0$ it yields $\forall x\varphi$. For $k \geq 1$ map it to the σ_φ -chain “cite σ_φ with $z := t_1$, then eliminate with $t_2, \dots, t_k$ ”, of probability $c^{k-1}(1-c) \prod_j Q(t_j)$. The conclusions coincide, since $\forall E$ with t_1 gives $\varphi(t_1) = \sigma_\varphi[z := t_1]$ (t_1 has no bound variable) and the remaining steps agree; failing chains map to failing chains. The map is a bijection onto all σ_φ -chains, and no σ_φ -chain yields $\forall x\varphi$ (its conclusions have fewer quantifiers). Summing over chains with conclusion s gives the factor c . Citation factorises over axioms, $\mu_T = \sum_A w_A \mu_A$, which gives the background form. The leading quantifiers of φ are rigid, so exactly $m+1$ chain lengths succeed from σ_φ and $m+2$ from $\forall x\varphi$, which gives the normalisers. For k variables the first k eliminations are absorbed into one citation. $\square$

c1 checks the identity on every output of $2 \cdot 10^5$ sampled derivations ($0 + x = x$, $\neg Sx = 0$, $\forall y(x + y = y + x)$; Q_{num} , Q_{GW}) to $5 \cdot 10^{-14}$; referee r1 to $1.6 \cdot 10^{-15}$.

Proof of Theorem 1.3. Each row follows from Proposition 1.2. **L0:** $H_\forall$ cites only $\forall x\varphi$. **L0^{cl}:** identical laws by definition. μ_T : factor c per datum. **L1:** divide by the normalisers; for quantifier-free φ , $Z_{H_{\text{sch}}} = 1 - c$ and the factor is $c(1-c)/((1-c) + c(1-c)) = c/(1+c)$. **L1^{sel}:** $\mu_{H_\forall}(s)/\mu_{H_\forall}(S) = c\mu_{H_{\text{sch}}}(s)/(c\mu_{H_{\text{sch}}}(S))$ as $\forall x\varphi \notin S$. **L1^{max}:** best chains map to best chains, each scaled by c . Scores: both theories prove every instance and, when $\forall x\varphi$ is consistent, are consistent with the data (if it is not, $S_{\text{prove}}(H_\forall) = 0$, which still agrees with Theorem 1.4). Background: $r(s) = \frac{(1-w)\mu_B(s) + wc\mu_{H_{\text{sch}}}(s)}{(1-w)\mu_B(s) + w\mu_{H_{\text{sch}}}(s)} \in [c, 1]$. $\square$

Proof of Theorem 1.4. (B1) is Theorem 1.3; (B3) is Proposition A.1 with the score argument above. (B2): fix w and $s \neq \forall x\varphi$. Under μ_T , $\mu_{B \oplus w \forall}(s) = (1-w)\mu_B(s) + wc\mu_{H_{\text{sch}}}(s) \leq (1-w)\mu_B(s) + w\mu_{H_{\text{sch}}}(s) = \mu_{B \oplus w \sigma}(s)$. Under **L1**, $Z_{B \oplus w \forall} = (1-w)Z_B + w((1-c) + cZ_{H_{\text{sch}}}) \geq (1-w)Z_B + wZ_{H_{\text{sch}}} = Z_{B \oplus w \sigma}$, since $(1-c) + cZ_{H_{\text{sch}}} - Z_{H_{\text{sch}}} = (1-c)(1 - Z_{H_{\text{sch}}}) \geq 0$; so the normalised likelihoods obey the same inequality datum by datum. A pointwise inequality at every w survives integration against a common weight prior. Under **L0^{cl}** the laws coincide at every w . $\square$

Proof of Proposition 1.5. (U2n) The unnormalised claim is the background row with $\mu_B := N$. For **L1**, $c\mu_{H_{\text{sch}}}(s)/Z_{H_\forall} \leq \mu_{H_{\text{sch}}}(s)/Z_{H_{\text{sch}}}$ because $Z_{H_\forall} = (1-c) + cZ_{H_{\text{sch}}} \geq Z_{H_{\text{sch}}}$ (as $Z_{H_{\text{sch}}} \leq 1$); mixing both with the same ηN keeps the inequality. Under **L1^{sel}** the selected laws coincide. A false datum has likelihood $\geq \eta N(s) > 0$. (U2t) The bijection maps a $\forall x\varphi$ -chain with $\ell+1$ eliminations to a σ_φ -chain with ℓ , with the same term product; $f(\ell+1)$ becomes $f(\ell)$. (U2c) H_{sch} ’s normalised law is Q_{σ_φ} for every c , and $c/(1+c)$ increases in c to $\frac{1}{2}$ at $c = 1$; a bound valid for every c survives integration. $\square$ c10(h) checked (U2n) at $2 \cdot 10^5$ random parameter points.

Proposition A.1 (Simulation inequality). *[proved]^{universal Prop U3} In a C_{U3} calculus, for quantifier-free φ and s , $\mu_{B \oplus \forall x\varphi}(s) - P^0(s) \leq c(\mu_{B \oplus \sigma_\varphi}(s) - P^0(s))$, with $P^0(s)$ the mass of derivations of s that never cite the added axiom.*

Proof of Proposition A.1. In a derivation of a quantifier-free s from $B \oplus \forall x\varphi$, a leaf citing $\forall x\varphi$ is not the root, and its parent takes a premise that is not quantifier-free; in a C_{U3} calculus only $\forall E$ does. So each such leaf sits in a pattern “ $\forall E(\text{cite } \forall x\varphi, t)$ ” concluding $\varphi(t)$. Replace each pattern by “cite σ_φ , $z := t$ ”: probability $cQ(t)p_{\text{cite}}w$ becomes $p_{\text{cite}}wQ(t)$, and conclusions are kept. The

map is injective, because cited axioms are labelled and every σ_φ -leaf in the image came from a pattern. Derivations with $m \geq 1$ patterns contribute $\sum c^m \Pr(f(\pi)) \leq c \sum \pi' \text{ citing } \sigma_\varphi \Pr(\pi')$; those with $m = 0$ never cite the added axiom and have the same probability in both theories. $\square$

The $\wedge$ -detour counterexample (Remark 1.6; proof sketch). $C_\wedge$ with probabilities p_c (cite), c ($\forall E$), a ($\wedge I$), e ($\wedge E$, side chosen with probability $\frac{1}{2}$); $B = \emptyset$; $\varphi = 0 + x = x$. A formula G made at a base node (cite or $\forall E$) reaches the root through a balanced word of $\wedge I$ (into a conjunction with a free partner) and $\wedge E$ (take G 's side), each pair of weight eaZ_T with Z_T the success probability. So $P_T(G) = \text{base}_T(G)D(eaZ_T)$ with $D(x) = \sum_j \text{Cat}_j x^j = (1 - \sqrt{1 - 4x})/(2x)$; the ansatz solves $P(G) = \text{base}(G) + e \sum_W P(G \wedge W)$ exactly when $D = 1 + xD^2$. For $H_\forall$, $\text{base}(\varphi(t)) = cP(\forall x\varphi)\mathcal{Q}(t)$ and $P(\forall x\varphi) = p_c D_\forall$, so $P_{H_\forall}(\varphi(t)) = cp_c D_\forall^2 \mathcal{Q}(t)$; for H_{sch} , $P(\varphi(t)) = p_c D_\sigma \mathcal{Q}(t)$. Z_T is the least fixed point of $Z = D(eaZ)(p_c + cP_U + aZ^2)$, $P_U = p_c D$ for $H_\forall$, 0 for H_{sch} ; the first map dominates, so $D_\forall \geq D_\sigma \geq 1$, with $D_\forall > 1$ if $a, e > 0$, and $R = cD_\forall^2/D_\sigma \geq cD_\forall > c$. Not written out: that the least fixed point is the success probability. At $(0.35, 0.15, 0.25, 0.25)$, $D_\forall = 1.03160$, $D_\sigma = 1.02636$, $R = 0.15553$ (c10(a), referee r2).

Proof of Proposition 1.7. Under $L1^{\text{sel}}$, T_σ gives $\varphi(t)$ probability $w\mathcal{Q}(t)$ and $\psi(t)$ probability $(1 - w)\mathcal{Q}(t)$. $T_\forall$ reaches S through $\forall x\varphi$ only after one elimination, so it gives $w_e\mathcal{Q}(t)$ and $(1 - w_e)\mathcal{Q}(t)$ with $w_e = wc/(1 - w + wc)$. The likelihoods are $K_D w_e^{n_\varphi} (1 - w_e)^{n_\psi}$ and $K_D w^{n_\varphi} (1 - w)^{n_\psi}$ with the same K_D . $w \mapsto w_e$ is a C^1 bijection of $[0, 1]$ with inverse $w = v/(c + v(1 - c))$ and $dw/dv = h(v)$, so

$$\text{BF}_n = \int_0^1 v^{n_\varphi} (1 - v)^{n_\psi} h(v) dv / \int_0^1 v^{n_\varphi} (1 - v)^{n_\psi} dv = \mathbb{E}[h(V)], \quad V \sim \text{Beta}(n_\varphi + 1, n_\psi + 1).$$

$n_\varphi/n \rightarrow w^*$ a.s. and the Beta variance is $\leq 1/(4(n + 3))$, so $V \rightarrow w^*$ in law; h is bounded and continuous. $h(v) > 1$ iff $\sqrt{c} > c + v(1 - c)$ iff $v < \sqrt{c}/(1 + \sqrt{c})$. $\square$

c10(b) ($c = 0.3$, 10 seeds, $n = 10^5$): 2.1918, 1.1537, 0.7102, 0.4056 at $w^* = 0.1, 0.3, 0.5, 0.8$, against $h(w^*) = 2.1914, 1.1534, 0.7101, 0.4056$.

Proof of Remark 1.8. Under $L1_{\text{cit}}^{\text{sel}}$ each citation is repeated until its chain lands in S . The component $\forall x\varphi$ lands only after one elimination, so its selected law is $\mathcal{Q}_{\sigma_\varphi}$, as is that of σ_φ ; σ_ψ has $\mathcal{Q}_{\sigma_\psi}$. Both theories therefore have law $(1 - w)\mathcal{Q}_{\sigma_\psi} + w\mathcal{Q}_{\sigma_\varphi}$ at every w . $\square$

Proof of Proposition 1.9. H_{both} with weight w on $\forall x\varphi$ has $\mu(\varphi(t)) = (1 - w + wc)(1 - c)\mathcal{Q}(t)$ and $Z = (1 - c)(1 + wc)$, so its ratio to H_{sch} is $(1 - w(1 - c))/(1 + wc)$. Since $1 - w + wc - (1 - w)(1 + wc) = w^2c \geq 0$, $I_n \geq \int_0^1 (1 - w)^n dw = \frac{1}{n+1}$; and the integrand is $\leq e^{-nw(1-c)}$, so $I_n \leq \frac{1}{n(1-c)}$. $\square$ Computed (c5 Part C): $nI_n = 0.958, 0.996, 1.000$ at $n = 10, 10^2, 10^5$.

A.2 Size principle, over-specific templates and memorisers

Proof of Theorem 1.10. (1) Log-sum inequality: $\sum_{s \in I} P^*(s) \ln \frac{P^*(s)}{P'(s)} \geq P^*(I) \ln \frac{P^*(I)}{P'(I)} = \ln \frac{1}{p}$. (2) With $Y_i = \ln \frac{P'(X_i)}{P^*(X_i)}$, $y^+ \leq e^y$ gives $\mathbb{E}Y^+ \leq \mathbb{E}e^Y = P'(\text{supp } P^*) \leq 1$, so $\mathbb{E}Y = -\text{KL} \in [-\infty, 0)$ and the strong law applies (it allows $\mathbb{E}Y = -\infty$ when $\mathbb{E}Y^+ < \infty$). (3) Tonelli: $\mathbb{E} \prod_i P'(X_i)/P^*(X_i) = P'(\text{supp } P^*)^n$. (4) Write $\sigma_\varphi = \tau\rho$, ρ not a renaming. Either (a) some metavariable u of τ has $\rho(u)$ with a rigid root f , or (b) ρ identifies two metavariables $u \neq u'$; otherwise ρ would be a renaming. If $\tau\theta \in \text{inst}(\sigma_\varphi)$, then $\tau\theta = \tau\rho\theta'$ and unique matching gives $\theta = \rho\theta'$

likelihood	hypothesis	$n = 0$	1	5	20	200
$L0^{cl}$ (= $L1^{sel}$ here)	$H_V; H_{sch}$	.021; .042	.194; .387	.303; .605	.332; .665	.332; .665
	over-general; over-specific	.926; .011	.307; .110	.022; .067	.000; .000	.000; .000
	root split, weights p_f	.000	.002	.002	.003	.003
μ_T	$H_V; H_{sch}$	.021; .042	.067; .448	.001; .891	.000; .996	.000; .996
	root split, weights p_f	.000	.002	.003	.004	.004
$L1$	$H_V; H_{sch}$	.021; .042	.053; .455	.000; .892	.000; .996	.000; .996

Table 3: Mean posterior over 50 seeds (c2): $\varphi = 0 + x = x$, $\mathcal{Q}_{num}$ ($q = \frac{1}{2}$), $\mathcal{C}_{min}$ ($c = 0.3$), prior $2^{-(symbols+1 \text{ per axiom})}$, $\alpha = 1$ for the learned-weight root split. Class: H_V , H_{sch} , four over-general templates ($?y = ?z$, $?y + ?z = ?z$, $0 + ?y = ?z$, and the bare formula metavariable, inconsistent, 0.674 at $n = 0$), $\varphi(Sz)$, and the root split with fixed and learned weights. Under $L0^{cl}$, $H_V : H_{sch}$ stays at the prior ratio $\frac{1}{2}$; under μ_T , H_{sch} saturates at 0.996 because the fixed-weight root split has exactly its law ($??$). Four other formulas and $\mathcal{Q}_{GW}$ behave alike.

on τ 's metavariables. In case (a) $\theta(u)$ has root f (probability $\mathcal{Q}(\text{root} = f)$, or $F(\text{root} = f)$); in case (b) $\theta(u) = \theta(u')$, of probability $\sum_t \mathcal{Q}(t)^2 \leq \max_t \mathcal{Q}(t) \leq \max_f \mathcal{Q}(\text{root} = f)$. $\square$

c2: under $\mathcal{Q}_{num}$, $?y + ?z = ?z$ has $P_\tau(I) = \frac{1}{2}$ and KL 0.693, $0 + ?y = ?z$ has $\frac{1}{3}$ and 1.386; (4) held in 38 of 38 cases and in the referee's 130 random generalisations (r7).

Proof of Proposition 1.11. $\mathcal{Q}(f(\bar{t})) = p_f \prod_i \mathcal{Q}(t_i)$ and τ matches $\varphi(t)$ iff $\text{root}(t) = f$, with $z_i := t_i$; so $P_\tau(\varphi(t)) = \mathcal{Q}(t)/p_f$ there and 0 elsewhere. Under H_{sch} 's law the ratio has mean $\sum_{\text{root}(t)=f} \mathcal{Q}(t)/p_f = 1$. $\square$ c2: gain per datum $\ln 2$ under $\mathcal{Q}_{num}$ and $\ln(1/0.3) = 1.204$ under $\mathcal{Q}_{GW}$; over-specific mass 0.110 at $n = 1$, 0 at $n = 10$ (Table 3).

Proof of Proposition 1.12. (i) $\pi(F) = Z_0 \prod_{s \in F} r_s / (1 - r_s)$. $F = S_n$ gives the lower bound $(r_s / (1 - r_s) \geq r_s)$. For the upper, $\sum_{F \supseteq S_n} \pi(F) = \prod_{s \in S_n} r_s$, and each likelihood is at most that of S_n : $|F|^{-n} \leq m_n^{-n}$, and j unused components multiply DirMult_α by $\frac{\Gamma((m+j)\alpha)\Gamma(n+m\alpha)}{\Gamma(m\alpha)\Gamma(n+(m+j)\alpha)} \leq 1$. (ii) The log-odds are at most a constant $-n \ln m_n + \sum_i \ln \frac{1}{\mathcal{Q}(t_i)}$, the sum is $n(H(\mathcal{Q}) + o(1))$ a.s., and $m_n \rightarrow \infty$ a.s. (ii') $\text{DirMult}_1(D_n) = 1 / [\binom{n}{(n_s)} \binom{n+m-1}{m-1}]$; the multinomial coefficient is $\leq 1/\text{ML}(D_n)$ (a type class has probability ≤ 1 under the maximum-likelihood law), $\binom{n+m-1}{m-1} \leq (n+m-1)^{m-1}$, and $\text{ML}(D_n) \geq P_{H_{sch}}(D_n)$; with the lower bracket this is the deterministic bound. For numerals $|\varphi(S^k 0)| = |\sigma_\varphi| + \text{occ} \cdot k$. With K_n the largest k seen, $P(K_n \geq 3 \log_{1/q} n) \leq n \cdot n^{-3} = n^{-2}$, summable, so by Borel–Cantelli a.s. $K_n < 3 \log_{1/q} n$ eventually; then $m_n \leq K_n + 1$ and $\sum_{s \in S_n} (|s| + 1) = O(\ln^2 n)$. $\square$

(The first version had probability $1 - 1/n$ at threshold $2 \log_{1/q} n$; c10(g) checks the n^{-2} tail exactly.) The universe must be named: over all closed sentences $\sum r_s = \infty$ at $\lambda = 1$ (referee r6). c3 ($\alpha = 1$, 10 seeds, log-odds up to a constant in $[\ln Z_0, 0]$): under $\mathcal{Q}_{num}$, $q = \frac{1}{2}$, $m_n = 10.4, 16.4$ at $n = 10^3, 10^5$ and the Laplace log-odds are $-133.8, -327.7$ ($-2.80 \ln^2 n, -2.47 \ln^2 n$; uniform weights $-1062, -1.4 \cdot 10^5$); at $q = 0.9$, $D_l / \ln^2 n$ moves from -35 to -53 (pre-asymptotic); under $\mathcal{Q}_{GW}$, $m_n = 2.2 \cdot 10^4$ at $n = 10^5$ and $D_l / n \approx -4.0$: exponential.

A.3 Predictive confirmation

Proof of Theorem 1.13. (1) By the chain rule, $\sum_{n < N} \mathbb{E} \text{KL}(P^* \| M(\cdot \mid D_n)) = \mathbb{E} \ln \frac{P^*(D_N)}{M(D_N)} \leq \ln \frac{1}{\pi(T^*)}$, as $M \geq \pi(T^*)P^*$. Pushing forward to the indicator of I , $\text{KL}(P^* \| M(\cdot \mid D_n)) \geq$

$\ln \frac{1}{M(I|D_n)} \geq 1 - M(I | D_n)$; a nonnegative series with finite mean is finite a.s. (2) $P_T(I)^\infty = 1[T \in G]$, and $C^* \subseteq G$, so ?? applies. (3) $1 - \pi_n(G) \leq \min(1, \sum_{T \notin G} \pi_n(T)/\pi_n(T^*))$ and Theorem 1.10(3). (4) $P_T(D_n) = (1 - \varepsilon_T)^n \hat{P}(D_n)$ on instance sequences. $\square$

c4: with a point mass $\pi_0 = 0.01$ and $\text{Uniform}(0, 1)$ on ε , $P(\text{all future} | n) = \pi_0/(\pi_0 + \frac{1-\pi_0}{n+1})$; with $\pi_0 = 0$ it is 0 while $P(\text{next } m) = \frac{n+1}{n+m+1}$. Summed misses 4.22 (Laplace) and 2.70 (dyadic) $\leq \ln 100 = 4.61$.

A.4 The Bayesian ω -gap

Proof of Theorem 1.15. (a) Sum the limit of ?? over $\text{Prov}_\forall$, or over the theories counted by Bel, Dis, Indep or Inc. (b) Both theories lie in C^* , one in $\text{Prov}_\forall$. For the learned-weight pair the marginal laws are exchangeable, not equal, so (a) does not apply; Proposition 1.7 gives the limit $\pi(T_\forall)h/(\pi(T_\forall)h + \pi(T_\sigma)) \in (0, 1)$ directly. (Under L1^{sel} , $\{\sigma_\varphi, \exists x \neg(0 + x = x)\}$ with S excluding the second sentence is a consistent refuter in C^* , so all three trichotomy values can keep mass.) (c1) $T \in C^*$ implies $\text{Th}_C(T) \cap X = \text{Th}_C(T^*) \cap X$. (c2) Each axiom instance of $T \in C^*$ lies in $\text{Th}_C(T^*) \cap X$, hence follows from T^* by soundness; symmetrically. (In $C_{\min}$, $T^* = \{\forall x(\varphi \wedge \psi)\}$ has limit 0 in (c1) and 1 in (c2): different questions, conflated in the first version.) (c3) In $C_{\min}$ every chain has positive probability under full-support $\mathcal{Q}$, so L1 is faithful, and a σ_φ -chain with quantifier-free φ can only cite. For the second part apply the rule to a derivation of an instance. (d1) $\mu_{H_\forall}(\cdot | S) = c\mu_{H_{\text{sch}}}(\cdot \cap S)/(c\mu_{H_{\text{sch}}}(S)) = P_{H_{\text{sch}}}$ since $\text{supp } \mu_{H_{\text{sch}}} = I_c \subseteq S$; then (c3) with $T^* := H_{\text{sch}}$, or (b) under L1^{sel} . (d2) For quantified φ , $\hat{P} := \mu_{H_{\text{sch}}}(\cdot | S)$ is the law of neither hypothesis. On S , $\mu_{H_\forall} = c\mu_{H_{\text{sch}}}$, so both log-ratios are constant there and $\text{KL}(\hat{P} \| P_{H_{\text{sch}}}) = \ln \frac{Z_{H_{\text{sch}}}}{\mu_{H_{\text{sch}}}(S)} < \ln \frac{Z_{H_\forall}}{c\mu_{H_{\text{sch}}}(S)} = \text{KL}(\hat{P} \| P_{H_\forall})$, because $cZ_{H_{\text{sch}}} < Z_{H_\forall}$; the posterior concentrates on H_{sch} (??). For a general class this is a sketch. (e) Pairs form a countable class of i.i.d. laws. (H_{sch}, S) has the data law iff $S \supseteq I_c$; $(H_\forall, S)$ has it iff $S \supseteq I_c$ and $\forall x \varphi \notin S$ (Theorem 1.3); otherwise some instance gets probability 0 or $\forall x \varphi$ positive probability. Summing prior masses over C^* gives the limit. $\square$

c10(e): filters {all sentences, closed quantifier-free, closed instances, closed instances with S -rooted argument}, $\pi = (\frac{1}{3}, \frac{2}{3})$, ν uniform: 0.333, 0.284, 0.252, 0.250 at $n = 0, 1, 5, 10$, predicted 0.2500.

stream	likelihood	class	$n = 0$	10	50	100
closed instances	L1	full	0.750	0.364	0.004	0.000
closed instances	L1	$\{H_\forall, H_{\text{open}}\}$	1.000	1.000	1.000	1.000
closed instances	L1^{sel}	full	0.750	0.750	0.750	0.750
output of H_{open}	L1	full	0.750	0.860	1.000	1.000
output of $H_\forall$	L1	full	0.750	1.000	1.000	1.000

Table 4: $P(T \vdash \forall x \varphi | D_n)$ in C_{open} (c5 Part A); closed instances are also $H_\forall$'s output filtered to closed instances; $\varphi = 0 + x = x$, defaults, prior $2^{-(\text{symbols}+1 \text{ per axiom})}$; full class $\{H_\forall, H_{\text{open}}, H_{\text{sch}} \text{ (guarded)}, H_{\text{both}}\}$, with a uniform prior on H_{both} 's weight, prior share of provers 0.750. On the full output of $H_\forall$ (itself reported), $H_\forall$ has 0.993 at $n = 10$.

Proof of Proposition 1.16. Under S_{prove} the posterior is the prior restricted to $\text{Adm}_n = \{T \text{ consistent} : T \vdash D_n\}$; every closed instance eventually appears a.s., so $\text{Adm}_n \downarrow \{T \text{ consistent} : T \vdash I_c(\varphi)\}$, and continuity of measure gives convergence in total variation. The model of AS

prior code	bits($H_\forall$)	bits(H_{sch})	share of $H_\forall$
universal: dtrc symbols +1 per axiom, $\lambda = 1$	7	6	0.333
model π_1 (??; 4 bits per token)	27	26	0.333
model π_2			0.200
experiments' template code, closed guard	17.18	17.09	0.484
pa, $\beta = 5$ bits per symbol	30	25	0.030
pa, $\beta = \log_2 23$	27.14	22.62	0.042

Table 5: Prior share of $H_\forall$ in $\{H_\forall, H_{\text{sch}}\}$ for $\varphi = 0 + x = x$ under the four tracks' codes (c10(f); the model rows by hand from ??). The experiments' value equals their E1 limit under $\text{L1}_{\text{cit}}^{\text{sel universal}} \S 6.6$.

Ex 4.9 shows that $\mathbf{Q}$, $\mathbf{Q} + \forall x\varphi$ and $\mathbf{Q} + \exists x\neg\varphi$ are consistent with all closed instances, each of which $\mathbf{Q}$ proves; $\{0 + z_1 = z_2\}$ has a one-element model. Under S_{nc} (and S_g , $g \equiv 1$) all theories proving the data get the same factor. Inc = 0 under the scores by definition. $\square$

φ , law, likelihood	$n = 0$	1	2	5	10	20
$0 + x = x$, $\mathcal{Q}_{\text{num}}$; L0^{cl} , L1 , L1^{sel}	0.674	0.000	0.000	0.000	0.000	0.000
$\neg Sx = 0$, $\mathcal{Q}_{\text{num}}$; L0^{cl} / L1	0.931	.585/.635	.319/.390	.056/.081	.002/.003	0.000
$x < Sx$, $\mathcal{Q}_{\text{num}}$; L0^{cl}	0.627	0.080	0.004	0.000	0.000	0.000
$\neg Sx = 0$, $\mathcal{Q}_{\text{GW}}$; L0^{cl}	0.931	0.454	0.191	0.010	0.000	0.000

Table 6: Mass Inc of inconsistent theories in the class of Table 3 (c10(d), 20 seeds): the bare formula metavariable and, for $\neg Sx = 0$, three templates with instances such as $\neg(0 = 0)$. $\text{Bel}_{\text{cons}}(\forall x\varphi)$ tends to the prior share ≈ 0.332 under L0^{cl} , L1^{sel} and to 0 under L1 .

Proof of Proposition 1.17. On $\mathbb{N} \cup \{a, b\}$, standard on $\mathbb{N}$, let $Sa = a$, $Sb = b$; $a + m = m$, $b + m = b$ ($m \in \mathbb{N}$); $x + a = a$ for $x \in \mathbb{N} \cup \{a\}$, $b + a = b$, $x + b = b$ for all x ; $x \cdot 0 = 0$; $a \cdot m = a$, $b \cdot m = b$ ($m \geq 1$); $n \cdot a = n \cdot b = b$ ($n \geq 1$); $0 \cdot a = 0 \cdot b = 0$; $a \cdot a = a \cdot b = a$; $b \cdot a = b \cdot b = b$. Q1–Q3: S is the identity on $\{a, b\}$ and $n \mapsto n + 1$ on $\mathbb{N}$. Q5: for $y = m$, $a + (m + 1) = m + 1 = S(a + m)$, both sides are b for $x = b$, and $x \in \mathbb{N}$ is standard; for $y \in \{a, b\}$, $Sy = y$ and $x + y \in \{a, b\}$ is fixed by S . Q6 holds by definition. Q7: for $y = m$, $a \cdot (m + 1) = a = a \cdot m + a$ (as $0 + a = a + a = a$) and $b \cdot (m + 1) = b = b \cdot m + b$; for $y \in \{a, b\}$ we need $z + x = z$ with $z = x \cdot y$, which holds for $x = 0$ ($0 + 0 = 0$), $x = n \geq 1$ ($b + n = b$), $x = a$ ($a + a = a$) and $x = b$ ($b + b = b$). Closed terms denote standard numbers, so every closed instance holds, but $a + 0 = 0 \neq a$. $\square$

Each case was re-checked for this paper; c6 and referee r5 find no violation on $\{0..40\} \cup \{a, b\}$ and $\{0..80\} \cup \{a, b\}$.

Proof of Proposition 1.18. $\forall x\varphi$ entails each instance, so $P(\forall x\varphi \wedge \bigwedge_{i \leq n} \varphi(t_i)) = P(\forall x\varphi)$, and the conjunctions decrease to “all closed instances hold”; continuity of μ gives the limit. The gap in the Gaifman condition is $\mu\{M : \text{all closed instances hold, } M \not\models \forall x\varphi\}$, which vanishes for every φ (countably many) iff μ -a.e. M satisfies (i) of AS Prop 4.7, i.e. $M_0 \preceq M$. $\square$

A.5 Open instances, guards and root splits

Proof of Proposition 1.19. (a) $P_T(D) > 0$ gives $T \vdash \varphi(p)$, and p occurs in no axiom, so Gen gives $T \vdash \forall x\varphi$. (b) follows from (a). (c) With weights $(w_\forall, w_{\text{open}}, w_{\text{sch}})$ on $\forall x\varphi$ and the

open- and closed-guarded σ_φ , the only outputs are $\forall x\varphi$, $\varphi(p)$ and closed $\varphi(t)$ (Gen fails on closed formulas, $\forall E$ on quantifier-free ones). With $a := \mu(\forall x\varphi)$, $b := \mu(\varphi(p))$: $a = p_{\text{cite}}w_\forall + gb$ and $b = \rho(p_{\text{cite}}w_{\text{open}} + ca)$, so

$$a = \frac{p_{\text{cite}}(w_\forall + g\rho w_{\text{open}})}{1 - g\rho}, \quad Z = a(1 + c) + p_{\text{cite}}(w_{\text{open}} + w_{\text{sch}}),$$

$$\mu(\varphi(t)) = \mathcal{Q}_{\text{num}}(t)[(1 - \rho)(p_{\text{cite}}w_{\text{open}} + ca) + p_{\text{cite}}w_{\text{sch}}] \quad (t \text{ closed}).$$

Hence $Z_{H_{\text{open}}} = p_{\text{cite}}\frac{1+g\rho}{1-g\rho}$, $Z_{H_\forall} = p_{\text{cite}}\frac{1+c}{1-g\rho}$, $Z_{H_{\text{sch}}} = p_{\text{cite}}$; unnormalised, $\mu_{H_\forall}/\mu_{H_{\text{open}}} = c$ at instances and $1/(g\rho)$ at $\forall x\varphi$, and $\mu_{H_{\text{open}}}/\mu_{H_{\text{sch}}} = \frac{1-\rho}{1-g\rho}$ at closed instances; divide by the normalisers. (d) follows from (c), Theorem 1.10(2) and Proposition 1.9. $\square$

c1 (sampler, $\max |z| = 1.60$), c5 and referee r7 (2.15) confirm (c).

Lemma A.2 (Waste). *[proved]^{universal Lemma N0} Let P^* be a law on a countable set, $H \cup T$ a partition of $\text{supp } P^*$ with $\tau := P^*(T) \in (0, 1]$, N_W a law on a set disjoint from $\text{supp } P^*$, and $r_w > 0$. Over the laws $p_h P^*(\cdot | H) + p_t P^*(\cdot | T) + r_w p_t N_W$ with $p_h + (1 + r_w)p_t = 1$, the minimum of $\text{KL}(P^* \parallel \cdot)$ is $\tau \ln(1 + r_w)$, attained at $p_t = \tau/(1 + r_w)$, $p_h = 1 - \tau$.*

Proof. The conditionals are exact and N_W lives off the support, so $\text{KL} = (1 - \tau) \ln \frac{1-\tau}{p_h} + \tau \ln \frac{\tau}{p_t}$, convex in p_t once $p_h = 1 - (1 + r_w)p_t$. The derivative $(1 - \tau)(1 + r_w)/p_h - \tau/p_t$ vanishes at $p_t = \tau/(1 + r_w)$, $p_h = 1 - \tau$, where the value is $\tau \ln(1 + r_w)$. $\square$

Proposition A.3 (Exact law of R_k). *[proved; computed]^{universal Prop N1} In $\mathcal{C}_{\text{open}}$, let $A_\rho := \frac{1-\rho}{1-g\rho}$ and $B_\rho := \frac{1+g\rho}{1-g\rho}$, and give R_k sentence weights $v_0, \dots, v_{k-1}$ and template weight u . Its unnormalised law is $p_{\text{cite}}v_j$ at $\varphi(S^j0)$ ($j < k$); $p_{\text{cite}}uA_\rho \mathcal{Q}_{\text{num}}(S^m0)$ at $\varphi(S^{k+m}0)$; $p_{\text{cite}}u\rho/(1 - g\rho)$ at $\varphi(S^k p)$; $p_{\text{cite}}u g\rho/(1 - g\rho)$ at $\forall y\varphi(S^k y)$; and 0 elsewhere. $Z = p_{\text{cite}}(1 - u + uB_\rho)$. H_{open} is the case $k = 0$, $u = 1$.*

Proof. Citing a sentence yields it, and Gen and $\forall E$ then fail. Citing the template yields $\varphi(S^k t)$, $t \sim \mathcal{Q}_{\text{open}}$; Gen applies only to $\varphi(S^k p)$, giving $U := \forall y\varphi(S^k y)$, and $\forall E$ only to U . With $a := \mu(U)$, $b := \mu(\varphi(S^k p))$: $a = gb$, $b = \rho(p_{\text{cite}}u + ca)$, so $a = g\rho p_{\text{cite}}u/(1 - g\rho)$, $b = \rho p_{\text{cite}}u/(1 - g\rho)$, and a closed tail instance has mass $(1 - \rho)\mathcal{Q}_{\text{num}}(S^m0)(p_{\text{cite}}u + ca) = p_{\text{cite}}uA_\rho \mathcal{Q}_{\text{num}}(S^m0)$. Summing gives $Z = p_{\text{cite}}(1 - u) + p_{\text{cite}}u(1 + g\rho)/(1 - g\rho)$. $\square$

c8 Part 1: sampler for $R_1..R_3$, $\max |z| = 2.07$ (referee r3: 2.19).

Proof of Proposition 1.21. Conditioned on closed outputs, $H_\forall$ and H_{open} have law P^* (Proposition 1.19(c)), so their KL is $-\ln$ of their closed mass, $c(1 - \rho)/(1 + c)$ and $A_\rho/B_\rho = (1 - \rho)/(1 + g\rho)$. For R_k (Proposition A.3) the normalised law has head mass $(1 - u)/Z_R$, $Z_R = 1 - u + uB_\rho$, tail mass $p_t = A_\rho u/Z_R$ with the exact tail shape (because $\mathcal{Q}_{\text{num}}(S^{k+m}0) = q^k \mathcal{Q}_{\text{num}}(S^m0)$), and waste $(B_\rho - A_\rho)u/Z_R = r_w p_t$ with $r_w = B_\rho/A_\rho - 1$. For fixed u the best head is $v_j \propto \mathcal{Q}_{\text{num}}(S^j0)$ (Gibbs), and Lemma A.2 applies with $\tau = q^k$: p_t ranges over $(0, A_\rho/B_\rho] = (0, 1/(1 + r_w)]$, which contains the optimum, attained at $u = q^k/(B_\rho(1 - q^k) + q^k)$. The value is $q^k \ln(B_\rho/A_\rho) = q^k \mathcal{W}_0$. $\square$

c8 Part 2: numerical minimisation agrees to $1.8 \cdot 10^{-15}$ ($k \leq 6$, four settings).

Proof of Proposition 1.22. *Provability.* On $\mathbb{N} \cup \{e, e'\}$, standard on $\mathbb{N}$, with $Se = Se' = e'$ and $0 + e = 0 + e' = e'$, every $S^k x$ ($k \geq 1$) is a standard number $\geq k$ or e' , so every instance of $\varphi(S^k z)$, closed or under closure, and $\varphi(0), \dots, \varphi(S^{k-1}0)$ hold, while $0 + e \neq e$. Over Q: Q proves

each $0 + S^j 0 = S^j 0$ (Q4, Q5, induction on j outside the theory), and k uses of Q3 show that every x is one of $0, \dots, S^{k-1} 0$ or $S^k y$, where $\varphi(S^k p)$, read as $\forall y \varphi(S^k y)$, applies. Memorisers consist of closed instances, which Q proves, and $Q \not\vdash \forall x \varphi$ (AS Ex 4.9). (a) $H_\forall, H_{\text{open}}$ have fixed laws with $KL \geq \mathcal{W}_0$ (note $\ln \frac{1+c}{c(1-\rho)} \geq \mathcal{W}_0$), so their normalised log-likelihood ratio to P^* is $-nKL + o(n)$ a.s. For R_k with learned weights, take the interior optimum w_0 and an ε -neighbourhood N_ε on which the weights are bounded away from 0; there the per-datum log-likelihood ($\ln v_j - \ln Z_R$, or $\ln(A_\rho u) + \ln \mathcal{Q}_{\text{num}}(S^m 0) - \ln Z_R$, whose middle term does not depend on w) has w -gradient bounded by some M . So the marginal is $\geq \text{Dir}(N_\varepsilon) P_{w_0}(D_n) e^{-nM\varepsilon}$, and $\liminf \frac{1}{n} \ln \frac{\text{marginal}}{P^*(D_n)} \geq -q^k \mathcal{W}_0 - M\varepsilon$ for every ε ; the log-odds of R_k against each prover grow like $n(1 - q^k) \mathcal{W}_0 - o(n)$. Provers that add σ_φ as a spare component to some R_k decay only polynomially relative to R_k (proof sketch; ??(b)). (b) All members but the memorisers prove $\forall x \varphi$ over Q, which gives the second sentence. By Proposition 1.12(ii') the memoriser class has log-marginal $\geq -C \ln^2 n$ relative to P^* . For each R_k the likelihood ratio to P^* depends only on the empirical frequencies $\hat{p}$ of the $k + 1$ categories (head values; tail), since the tail shape cancels; so $\frac{1}{n} \ln \sup_w \frac{P_{R_k, w}(D_n)}{P^*(D_n)} = \sup_w G(\hat{p}, w)$, convex and hence continuous in $\hat{p}$ near p^* , with limit $-q^k \mathcal{W}_0 < 0$, and the marginal is below the supremum. With $\mathcal{K}$ finite all non-memorisers lose linearly against the memorisers. (c) (Proof sketch.) Under $L1^{\text{sel}}$, R_k 's filtered law (head v_j/N , tail $A_\rho u \mathcal{Q}_{\text{num}}(S^m 0)/N$, $N = 1 - u + A_\rho u$) is a smooth reparametrisation of a $(k + 1)$ -category multinomial equal to P^* at one interior point. The Laplace (BIC) expansion (Schwarz, 1978; Clarke and Barron, 1990) (recalled, details not checked) gives log-marginal $-\frac{k}{2} \ln n + O_P(1)$ relative to P^* ; memorisers lose $O(\ln^2 n)$; $H_\forall, H_{\text{open}}$ have log-ratio 0. Not written out: almost-sure control, uniform in k for infinite $\mathcal{K}$. $\square$

likelihood, class	quantity	$n = 0$	10	10^2	10^3	10^4	10^5	10^6
L1, $\{H_\forall, H_{\text{open}}, R_1\}$	$P(\vdash)$, pure	.997	.996	.697	.000	.000	.000	.000
	MAP	H_{open}	H_{open}	H_{open}	R_1	R_1	R_1	R_1
L1, $+R_{2..4}$, mem.	$P(Q \cup T \vdash)$	.601	1.000	1.000	1.000	1.000	.000	.000
	MAP	H_{open}	H_{open}	H_{open}	R_3	R_4	mem	mem
L1, $+R_{2..40}$, mem.	$P(\vdash)$, pure	.599	.996	.697	.000	.000	.000	.000
	$P(Q \cup T \vdash)$	.601	1.000	1.000	1.000	1.000	1.000	1.000
	MAP	H_{open}	H_{open}	H_{open}	R_3	R_6	R_9	R_{12}
$L1^{\text{sel}}$, each class	$P(\vdash)$, pure	.997/.599	.998	.987	1.000	1.000	1.000	1.000

Table 7: Without a guard (c8 Part 3): $P(T \vdash \forall x \varphi \mid D_n)$ in pure logic and over Q, $\varphi = 0 + x = x$, closed numeral data ($q = \frac{1}{2}$), C_{open} at $(c, g, \rho) = (0.3, 0.2, 0.1)$, Dirichlet($\frac{1}{2}$) weights, prior $2^{-(\text{symbols}+1 \text{ per axiom})}$, memoriser class mass 2^{-6} (lower bracket), mean of 20 seeds; MAP = mode over seeds. Part 4 (multinomial counts, 10 seeds): with $R_{1..40}$ the MAP is $R_{15}, R_{18}, R_{21}, R_{24}, R_{30}$ at $n = 10^7, 10^8, 10^9, 10^{10}, 10^{12}$ and $P(Q \cup T \vdash) = 1.000$, $P(\vdash)$ pure = 0, also with the upper bracket and class mass $\frac{1}{2}$, with $\alpha = 1$, and with $\lambda = 2$ ($R_{14}..R_{30}$)^{universal Prop N3}.

The MAP depth is about $\log_2 n$ minus a slowly growing term (≈ 8 at $n = 10^6$, ≈ 10 at 10^{12}).

A.6 Quantified data and the equational fragment

Proof of Proposition 1.23. (a) If $B \cup I_c \not\vdash s$, then $B \oplus \sigma_\varphi \not\vdash s$. In AS Ex 4.9's model, $0 + Sa = 0 + a = b \neq a = Sa$, so $Q + I_c \not\vdash \forall y(0 + Sy = Sy)$; and $Q + \forall y(0 + Sy = Sy) \vdash \forall x(0 + x = x)$ by Q4 at 0 and Q3 elsewhere. (b) In units of $1 - c$, with $S_B := \sum_{k=0}^{L+1} c^k \cdot \{\forall x \varphi, \sigma_\varphi\}$ with weight w on $\forall x \varphi$ gives $P(\forall x \varphi) = \frac{w}{1+wc}$, $P(\varphi(t)) = \frac{1-w+wc}{1+wc} \mathcal{Q}(t)$; $\{\forall^L \forall x \varphi, \sigma_\varphi\}$ gives $P(\forall x \varphi) = wc^L / Z_B$,

$P(\varphi(t)) = ((1-w) + wc^{L+1})\mathcal{Q}(t)/Z_B$, $Z_B = wS_B + 1 - w$ (vacuous eliminations succeed with term mass 1). A quantified datum adds $L \ln \frac{1}{c} + \ln \frac{Z_B}{1+wc}$ to the log-ratio; an instance 0.021, 0.094, 0.307, 1.323 nats at $w = 0.05, 0.2, 0.5, 0.9$ ($L = 3$), so the two theories price instances alike only for small w (referee m5). For $f_q = 0$ both Bayes factors against H_{sch} are $\int_0^1 e^{nf_i(w)} dw$ with $f_i(0) = 0$, f_i decreasing, $f'_{\text{both}}(0) = -1$, $f'_B(0) = -(1+c+\dots+c^L)$; Watson’s lemma gives $(n|f'_i(0)|)^{-1}(1+o(1))$ for each. For $f_q > 0$ (sketch) the rate is $\Delta(f_q) := \max_w \mathbb{E}\ell_{\text{both}}(w) - \max_w \mathbb{E}\ell_B(w)$, the Laplace approximation at the interior maxima not written out. (c) (Sketch.) In $\mathbf{L1}^{\text{max}}$ form a derivation’s code length is a sum of rule and term costs; adding ψ costs its prior bits and an $O(\ln n)$ dilution, and saves $\ell_T(s) - \ell_{T+\psi}(s)$ steps of cost $\geq \ln \frac{1}{c}$ per datum s . $\square$

c10(c), equal priors, 20 seeds, $n \leq 10^5$: for $f_q = 0$ the log Bayes factor is 0.349 ($L = 3$) and 0.357 ($L = 10$) at every n ; for $L = 3$, $\Delta = 0.0396, 0.1980, 0.7921$ at $f_q = 0.01, 0.05, 0.2$ (measured 0.0395, 0.1977, 0.7921; $f_q L \ln \frac{1}{c} = 0.036, 0.181, 0.722$); for $L = 10$, 0.1240, 0.6198, 2.4793 (measured 0.1236, 0.6187, 2.4793). The first version’s table (c5 Part B) gave $\{\forall^L \forall x \varphi, \sigma_\varphi\}$ a much smaller prior and so tracked only the refutation in (a): $P(T \vdash \forall x \varphi) = 1 - (1 - f_q)^n$ within Monte Carlo error.

Proposition A.4 (The equational fragment). *[(a)–(c) proved; (a) computed]^{universal Prop U14} (a) In equational logic over $E = \{Q4, Q5\}$ every derivation of $0 + S^k 0 = S^k 0$ has at least $k + 1$ axiom leaves. (b) If grammar nodes pick rules independently, with at most two premises and mean $\mu < 1$, then $P(\geq L \text{ nodes}) \leq e^{-(L-1)(1-\mu)^2/2}$. (c) The full sum favours the schema by $\Omega(k)$ on $0 + S^k 0 = S^k 0$ when $q > e^{-(1-\mu)^2/2}$; this needs $q > 0.61$, so it does not apply at $q = \frac{1}{2}$; the two-part ($\mathbf{L1}^{\text{max}}$) comparison favours it linearly in k at $q = \frac{1}{2}$ when each leaf pays a citation and a term of point mass $\leq 1 - q$.*

Proof of Proposition A.4. (a) Let $w(t) := \sum_{+-\text{nodes } l+r} (1 + \#S(r))$, $\#S(r)$ the number of S symbols in r . Rewriting $u + 0$ to u , or $u + Sv$ to $S(u + v)$, anywhere in a term changes w by exactly -1 (the rewritten node contributes 1, resp. $2 + \#S(v)$ before and 0, resp. $1 + \#S(v)$ after; nodes inside are untouched; nodes above keep the S -count of their right argument), and the reverse by $+1$. $w(0 + S^k 0) = k + 1$, $w(S^k 0) = 0$. A derivation (reflexivity, symmetry, transitivity, congruence, substitution, axiom instances) converts by induction into a rewrite chain with at most as many steps as axiom leaves: congruence and transitivity concatenate chains, symmetry reverses one, substitution keeps lengths. (b) Couple the grammar with the Galton–Watson tree of rule labels and explore it depth-first with premise counts ξ_i ; $\geq L$ nodes requires $\sum_{i \leq L-1} (\xi_i - 1) \geq 0$; apply Hoeffding’s inequality (Hoeffding, 1963) to $\xi_i - 1 \in [-1, 1]$ with mean $\mu - 1$ (typed grammars need the maximum of μ over contexts). (c) Full sum: by (a), (b), $P_E(0 + S^k 0 = S^k 0) \leq e^{-k(1-\mu)^2/2}$, while the schema gives $\geq p_{\text{cite}} w(1 - q) q^k$. Two-part: $\geq k + 1$ leaves of cost $\geq \ln \frac{1}{p_{\text{cite}}} + \ln \frac{1}{1-q} > \ln \frac{1}{q}$ (for $q \geq \frac{1}{2}$), against $O(1) + k \ln \frac{1}{q}$ for the schema. $\square$

c6: shortest chains of exactly $k + 1$ for $k \leq 5$ by breadth-first search; referee r5: the invariant on 339 250 random rewrites.

A.7 The sentence-only class

Proof of Proposition 1.24. (a) (Sketch.) Split_k equals the true law at one weight vector ($v_j = \mathcal{Q}_{\text{num}}(S^j 0)$, $u = q^k$) and pays $n^{-k/2}$ with learned weights; $\text{Both} 0$ only at the boundary; memorisers lose $O(\ln^2 n)$; uniformity in k not written out. (b) $H_\forall$ gives each instance $c/(1 + c)$ times its true probability (Theorem 1.3). Split_k : the universal sentence yields itself with mass $(1 - c)u$ or $\varphi(S^{k+m} 0)$ with mass $(1 - c)uc\mathcal{Q}_{\text{num}}(S^m 0)$; normalised by $(1 - c)(1 + cu)$: head $v_j/(1 + cu)$, tail $uc\mathcal{Q}_{\text{num}}(S^m 0)/(1 + cu)$, waste $u/(1 + cu) = p_t/c$; Lemma A.2 with $\tau = q^k$,

$r_w = 1/c$ gives $q^k \ln(1 + \frac{1}{c})$, attainable as p_t ranges over $(0, \frac{c}{1+c}]$. Both0 with weight w on $\varphi(0)$: with $M := (1-w)c/(1+c(1-w))$, the tail $\{j \geq 1\}$ has mass qM with the exact shape, waste M/c , and $\varphi(0)$ mass $1 - qM - M/c$; so $\text{KL} = (1-q) \ln \frac{1-q}{1-M(q+1/c)} + q \ln \frac{1}{M}$, convex, minimal at $M = \frac{qc}{1+qc} \leq \frac{c}{1+c}$ with value $q \ln \frac{1+qc}{qc}$. (c) The countermodel and derivation of Proposition 1.22. $\square$

c9 Parts 1–2 confirm the rates to $8.9 \cdot 10^{-16}$; referee r4 reproduces them.

Proof of Lemma 1.25. A chain cites some $A \in T$ and eliminates with numerals; T is finite, so one A yields $\varphi(S^j 0)$ for infinitely many j . The conclusions are quantifier-free, so $A = \forall \bar{y} \chi$ with χ quantifier-free and $\chi[\bar{t}/\bar{y}] = (0 + S^j 0 = S^j 0)$ for numeral tuples with infinitely many j . Numerals contain only 0 and S , so χ is $\tau_0 + \tau_1 = \tau_2$ with τ_0 either 0 or a variable instantiated by 0 in every such output, and τ_1, τ_2 of the forms $S^a 0$ or $S^a y$. As j varies, $\tau_1 = S^a y_m$ and $\tau_2 = S^b y_{m'}$, and neither variable is that of τ_0 (which takes only the value 0). If $m = m'$, then $a = b$ and $A \vdash \forall y \varphi(S^a y)$ (instantiate τ_0 's variable, if any, by 0); over $\mathbf{Q}$, a uses of Q3 and the closed instances give $\forall x \varphi$. If $m \neq m'$, $A \vdash 0 + S^a 0 = S^b 0$ and $A \vdash 0 + S^a 0 = S^{b+1} 0$, so $S^b 0 = S^{b+1} 0$, which $\mathbf{Q}$ refutes; then $\mathbf{Q} \cup T$ is inconsistent. $\square$

The track's proof treats only $\tau_0 = 0$; the case of a variable always instantiated by 0 is added here and does not change the conclusion.

likelihood, class	quantity	$n = 0$	20	10^2	10^3	10^4	10^5	10^6
L1, $\{H_\forall, \text{Split}_1, \text{Both0}\}$	$P(\vdash)$, pure	0.996	0.038	0.000	0.000	0.000	0.000	0.000
L1, $+\text{Split}_{1..4}$, mem.	$P(\mathbf{Q} \cup T \vdash)$	0.338	0.926	0.947	0.475	0.000	0.000	0.000
	MAP	mem	Split ₁	Split ₃	mem	mem	mem	mem
L1, $+\text{Split}_{1..40}$, mem.	$P(\vdash)$, pure	0.336	0.024	0.000	0.000	0.000	0.000	0.000
	$P(\mathbf{Q} \cup T \vdash)$	0.338	0.926	0.947	1.000	1.000	1.000	1.000
	MAP	mem	Split ₁	Split ₃	Split ₆	Split ₉	Split ₁₂	Split ₁₅
L0 ^{cl} , $+\text{Split}_{1..40}$, mem.	$P(\vdash)$, pure	0.336	0.997	0.999	1.000	1.000	1.000	1.000

Table 8: Sentence-only class (c9 Part 3): $\varphi = 0 + x = x$, $C_{\min}$, $c = 0.3$, numerals with $q = \frac{1}{2}$, Dirichlet($\frac{1}{2}$), prior $2^{-(\text{symbols}+1 \text{ per axiom})}$, memoriser class mass 2^{-6} (lower bracket), 20 seeds, MAP = mode; the larger classes also contain $H_\forall$ and Both0. With Split_{1..40} under L1 the MAP is Split₁₈, Split₂₁, Split₂₄, Split₂₇, Split₃₄ at $n = 10^7, \dots, 10^{10}, 10^{12}$ with $P(\mathbf{Q} \cup T \vdash) = 1.000$, also with the upper bracket and class mass $\frac{1}{2}$, with $\alpha = 1$ and with $\lambda = 2$ (c9 Part 4).

The race (c9 Part 5). Log odds of the split family ($k \leq 40$) against the memoriser family, mean of 10 seeds: 16.5 nats at $n = 10^2$, 54.7 at 10^4 , 105.1 at 10^6 , 193.5 at 10^{10} and 286.4 at 10^{12} (minimum over seeds negative only at 10^2), of the order of $L \log_2 L$ with $L = \log_2 n$. *Heuristic (conjecture).* The best memoriser pays about L^2 bits of prior and $(L/2) \ln n$ nats of Dirichlet cost; the best split wastes about $nq^k \ln \frac{1+c}{c}$ nats, balancing the $\Theta(L)$ cost of one more level at $k \approx L - \log_2 L$, and so saves about $\log_2 L$ levels of $\Theta(L)$ nats. *History.* The first version's “= 1 over $\mathbf{Q}$ ” came from $\{H_\forall, \text{Split}_1, \text{Both0}\}$, where all members prove $\forall x \varphi$ over $\mathbf{Q}$; the referee's “ $\rightarrow 0$ over $\mathbf{Q}$ ” holds with depth ≤ 4 , not with depth ≤ 40 to $n = 10^{12}$.

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